Real-Weight Optimal-Class Kernels for Explicit LCHS Circuits: Fewer Gates at Matched Precision for Dissipative Quantum Simulation

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Abstract

The linear combination of Hamiltonian simulations (LCHS) writes a dissipative propagator e^{-At} as a kernel-weighted integral of unitary evolutions whose truncation radius sets the cost. The most compact explicit LCHS circuit—a sinusoidal (Fejér–Clenshaw–Curtis) quadrature whose 2^{n_k} Hamiltonian simulations share one quantum-signal-processing selector—uses the near-optimal An–Childs–Lin kernel, with radius $O(\log^{1/\beta}(1/\epsilon))$. We run Low and Somma's optimal-class Gaussian-damped kernel, with radius $O(\log(1/\epsilon))$, inside this circuit without changing its selector, quadrature, qubitisation or block-encoding. The enabling step is exact: the kernel's linear phase is absorbed into the existing block-encoding of kA_L by a diagonal shift that changes one rotation angle, after which all weights are real, even and positive and are prepared by one even-parity QSVT with exact amplitude amplification and the reference circuit's three weight ancillas. We show that the quadrature error is governed by the residue pole at $k = -\mathrm{i}$, so the precision reachable with a given k-register is set by the kernel's radius. In 174 GPU emulations of the advection–diffusion benchmark at matched accuracy and no larger LCU one-norm (hence matched success probability), the substituted kernel uses $1.86\times$ fewer gates at the headline configuration and 1.59–$3.13\times$ fewer over eleven targets in $[10^{-4}, 10^{-1.5}]$, scales linearly in $\|A\|t$ like the reference, and reaches precisions the reference cannot reach with a 6-qubit k-register. Across seven circuits from five kernel families, exact kernels are never cheapest; Low and Somma's rational kernel becomes cheaper per amplitude-amplified success below 10^{-3}. In our window the gain is a constant factor, largest at low precision: the $\log(1/\epsilon)$ exponent is not yet visible, and a problem-independent design from the approximate-LCHS bound shows the same, with radius ratios of 1.46–2.52.

1 Introduction

Linear dissipative dynamics $\partial_t \psi = -A\psi$ with a non-normal generator A is ubiquitous in transport, plasma physics and fluid mechanics, and it is also what remains after linear embeddings of nonlinear dynamics such as the Koopman–von Neumann and Carleman constructions [Joseph, 2020, Novikau and Joseph, 2025a]. Quantum computers apply unitaries, so the non-unitary propagator e^{-At} has to be encoded indirectly. The linear combination of Hamiltonian simulations (LCHS) [An et al., 2023, 2025a] does so by writing, for $A = A_L + \mathrm{i}A_H$ with Hermitian $A_L \succeq 0$ and A_H,

$$e^{-At} = \int_{\mathbb{R}} g(k)\, U(t;k)\, dk, \qquad U(t;k) = e^{-\mathrm{i}(A_H + kA_L)t}, \qquad g(k) = \frac{\xi(k)}{1 - \mathrm{i}k}, \tag{1}$$

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which turns the dissipative evolution into a linear combination of unitaries (LCU) [Childs and Wiebe, 2012]. Its cost is governed by the kernel ξ: the integral is truncated to $|k| \le k_{\max}$, and the most expensive Hamiltonian simulation in the combination costs $\Theta(k_{\max}\|A_L\|t)$ queries. Three kernel regimes are known. The Cauchy kernel of the original LCHS needs $k_{\max} = O(1/\epsilon_{\mathrm{LCHS}})$ [An et al., 2023]. The An–Childs–Lin (ACL) kernels satisfy (1) exactly and need $k_{\max} = O(\log^{1/\beta}(1/\epsilon_{\mathrm{LCHS}}))$, $\beta < 1$; no exact kernel can do better [An et al., 2025a]. Approximate kernels obtained by analytic extension of a smoothed exponential decay reach the optimal $k_{\max} = O(\log(1/\epsilon_{\mathrm{LCHS}}))$ [Low and Somma, 2025, Jin et al., 2025b, de Lima Silva et al., 2022].

Progress on kernels has outpaced explicit circuits. The most compact explicit LCHS circuit [Novikau and Joseph, 2025b] maps the integration variable to $k = k_{\max}\sin\theta$—equivalently, Fejér–Clenshaw–Curtis (FCC) quadrature—so that the $N_k = 2^{n_k}$ Hamiltonians $A_H + \sin\theta_j\, k_{\max}A_L$ are block-encoded *simultaneously* and simulated by a single quantum-signal-processing (QSP) circuit [Low and Chuang, 2017, 2019] with $O(1)$ ancillas besides the n_k-qubit k-register. That circuit uses the near-optimal ACL kernel with $\beta = 0.7$. Low and Somma (LS) give circuits for their optimal kernel [Low and Somma, 2025], but on a uniform k-grid and with a multiplexed block-encoding that requires additional registers. Whether an optimal-class kernel can run *inside* the FCC/single-QSP circuit without changing its structure, and whether it pays off at resolutions that can actually be executed on an emulator, has not been examined. This paper does so for the Gaussian-damped kernel $\hat{f}_2$ of LS.

Contributions.

1. *Exact phase absorption.* The analytic-extension kernel carries a linear phase $e^{-\mathrm{i}ck}$. On the FCC grid $k_j = k_{\max}\sin\theta_j$ is not linear in j, so this phase cannot be applied by single-qubit phase gates as on a uniform grid. We show that it can nevertheless be absorbed exactly into the block-encoding the FCC circuit already contains, by replacing $k_{\max}A_L$ with $k_{\max}(A_L + \frac{c}{t}I)$—one rotation angle changes. After absorption the LCHS weights are real, even and positive; within the LS family $\hat{f}_{j,y}$ this happens only for $\hat{f}_2 = \hat{f}_{2,1}$ (Sec. 3.2).

2. *One-QSVT weight preparation.* Real, even weights are prepared by a single even-parity quantum singular value transformation (QSVT) [Gilyén et al., 2019] on the same $\sin\theta$ block-encoding, with exact amplitude amplification so that both weight preparations share their flag qubits: three weight ancillas, as in the brute-force weight oracle emulated in Novikau and Joseph [2025b], and two fewer than a complex-weight QSVT design (Sec. 3.5). All angle sets are generated by QuCF's own angle finder and validated independently.

3. *A resolution law for FCC-based LCHS.* The FCC error is controlled by the pole at $k = -\mathrm{i}$ that every LCHS kernel must carry, $\sim e^{-2(N_k-1)\,\mathrm{asinh}(1/k_{\max})}$, so at a fixed k-register the attainable precision is set by the kernel's truncation radius (Sec. 3.4). This is why the kernel choice matters most at small n_k.

4. *Emulator evidence.* On the QuCF/QuEST GPU emulator [Novikau, 2024, Jones et al., 2019], for the advection–diffusion benchmark of Novikau and Joseph [2025b], at matched or better final-state accuracy and no larger LCU one-norm (hence matched success probability), the substituted kernel needs $1.86\times$ fewer single-target multi-controlled (STMC) gates at the headline configuration and 1.59–$3.13\times$ fewer over eleven precision targets in $[10^{-4}, 10^{-1.5}]$; the gate count grows linearly with $\|A\|t$ for both kernels, so the reduction is a prefactor at unchanged $\|At\|$-exponent (Sec. 5). We map which kernel family is cheapest at which $(n_k, \epsilon_{\mathrm{LCHS}})$ (24 cells, seven circuits from five kernel families) and report where the new kernel does *not* help: over the emulated window its advantage is a constant factor that is largest at low precision, so the fitted exponent of the gate count in $\log(1/\epsilon_{\mathrm{LCHS}})$ is not smaller than for the ACL kernel—the $O(\log 1/\epsilon_{\mathrm{LCHS}})$ asymptotics is not reached at $\epsilon_{\mathrm{LCHS}} \ge 10^{-4}$, also not in a problem-independent analysis of the worst-case bound, whose designs keep radius ratios of 1.46–2.52 against the emulated 1.8–3.7 (Sec. 6.2).

The selector, quadrature, qubitisation and block-encoding are held fixed throughout, so all differences are attributable to the kernel and its weight preparation. We do not attempt the Koopman–von Neumann benchmark here.

2 Background

Kernels. Equation (1) holds exactly whenever ξ is analytic in the lower half plane, decays there, and is normalised by $\int g\,dk = 1$ [An et al., 2025a]; closing the contour picks up the residue of $1/(1-\mathrm{i}k)$ at $k=-\mathrm{i}$, where $U(t;-\mathrm{i}) = e^{-At}$. The ACL family

$$\xi_\beta(k) = \left[2\pi e^{-2^\beta}\,\exp((1+\mathrm{i}k)^\beta)\right]^{-1}, \qquad \beta \in (0,1), \tag{2}$$

decays as $|\xi_\beta(k)| \sim e^{-\cos(\beta\pi/2)|k|^\beta}$; a no-go theorem forbids exact kernels with exponential decay [An et al., 2025a]. LS relax exactness [Low and Somma, 2025]: if $\hat{f}(z)$ decays uniformly in a strip $\mathrm{Im}\,z \in [-y_0, 0]$, $y_0 > 1$, and $(z+\mathrm{i})\hat{f}(z)$ is analytic there with $\mathrm{Res}_{z=-\mathrm{i}}\hat{f} = \mathrm{i}/\sqrt{2\pi}$, then with $g = \hat{f}/\sqrt{2\pi}$

$$\left\| \int_{-R}^{R} g(k)U(t;k)dk - e^{-At} \right\| \le \frac{1}{\sqrt{2\pi}}\left[\int_{|k|>R} |\hat{f}(k)|dk + \int_{\mathbb{R}} |\hat{f}(k-\mathrm{i}y_0)|dk \right] \tag{3}$$

(LS Theorem 1). The kernels that decay fastest are entire approximations of $e^{-|x|}$ in the conjugate variable; their Fourier transforms are what enters (1). The cost of a kernel is summarised by its truncation radius $R = k_{\max}$ (selector cost) and its LCU one-norm $\alpha = \int |g|$ (success probability $\propto \alpha^{-2}$); for the ACL kernel $\alpha_{0.7} = 1.305$, $\alpha_{0.9} = 2.01$ (quadrature, `logs/kernel_derivation`).

The FCC/single-QSP circuit. With $\theta_j = -\pi/2 + j\Delta\theta$, $\Delta\theta = \pi/(N_k-1)$, $k_j = k_{\max}\sin\theta_j$, the truncated integral becomes [Novikau and Joseph, 2025b]

$$\begin{aligned} e^{-At} &\approx \sum_{j=0}^{N_k-1} w_j V_j, \qquad w_j = k_{\max}\cos\theta_j\,\Delta\theta\,g(k_j), \\ V_j &= e^{-\mathrm{i}C_j t}, \qquad C_j = A_H + \sin\theta_j\,B_m, \qquad B_m = k_{\max}A_L. \end{aligned} \tag{4}$$

A SIN gate rotates one ancilla by an angle linear in j so that $\langle 0|\mathrm{SIN}|0\rangle = \mathrm{diag}_j(\sin\theta_j)$; an LCU of the block-encodings U_{A_H}, U_{B_m} and of SIN then block-encodes $\bigoplus_j C_j/(2\eta)$ for a common normalisation η, and a single fully coherent QSP circuit of this block-encoding applies V_j to every branch $|j\rangle$ at once (the *selector*). The weights enter through a state preparation $\sum_j \sqrt{w_j}\,|j\rangle$ before and after the selector (PREP†·SEL·PREP), with amplitude amplification (AA) of the preparation [Brassard et al., 2002]. In the reference implementation the preparation is a brute-force multiplexed R_yR_z rotation that hard-codes ξ_β (QuCF `DirDec` profile), with a separate flag qubit for the left and right preparations. For the advection–diffusion equation, A_H and A_L are two- and three-banded circulants block-encoded with incrementer/decrementer circuits.

Cost accounting. We count (i) STMC gates, the gate set QuCF compiles to, for the complete circuit (initial-state preparation and its AA, both weight preparations with AA, and the selector including every controlled block-encoding of A_H and A_L); (ii) Q_A, the number of controlled applications of the block-encoding U_C or $U_C^\dagger$ (each containing one controlled U_{A_H} and one controlled U_{B_m}), read from a counter we added to QuCF's QSP constructor; (iii) ancillas, all qubits except the n_x-qubit state register (the k-register included); (iv) the success probability, i.e. the squared norm of the all-ancillas-zero output. The selector's QSP length is set by $\tau = 2\eta t$; with the reference normalisation $\eta \propto k_{\max}\|A_L\|$ the selector cost is $\Theta(k_{\max}\|A_L\|t)$ and dominates the circuit (99.8% of the STMC gates in the published $n_k = 7$ run, `runs/repro_check`).

3 Method: an optimal-class kernel inside the FCC/single-QSP circuit

3.1 The kernel

We use the $j = 2, y = 1$ member of the LS family [Low and Somma, 2025, Lemma 6, Cor. 7], written in the normalisation of (1):

$$\xi_{\gamma,c}(k) = \frac{e^{c(1-\mathrm{i}k)}\,e^{-(1+k^2)/(4\gamma^2)}}{\pi(1+\mathrm{i}k)}, \qquad g(k) = \frac{e^{c}\,e^{-\mathrm{i}ck}\,e^{-(1+k^2)/(4\gamma^2)}}{\pi(1+k^2)}, \qquad \gamma>0,\ c>0. \tag{5}$$

Its conjugate function is $f_2(x) = e^c[f_1(x+c) + f_1(-x-c)]$ with $f_1(y) = \frac{1}{2}e^{-y}\mathrm{erfc}(\frac{1}{2\gamma} - \gamma y)$: the cusp $e^{-|x+c|+c}$ smoothed by a Gaussian of width $\sim 1/\gamma$ and moved to $x = -c$, so that $f_2(x) \approx e^{-x}$ on the spectrum $x = \lambda t \geq 0$ of $A_L t$ with error $O(e^{c-\gamma^2 c^2})$. The kernel has a Gaussian tail, LCU one-norm $\alpha = e^c \,\mathrm{erfc}(1/2\gamma) \leq e^c$, and for fixed c the choice $\gamma = c^{-1}[c + \ln((1+1/2\pi)/\epsilon_{\mathrm{LCHS}})]^{1/2}$, $k_{\max} = 2c\gamma^2$ guarantees error $\leq \epsilon_{\mathrm{LCHS}}$ in (3), i.e. $k_{\max} = O(c^{-1}\log(1/\epsilon_{\mathrm{LCHS}}))$ [Low and Somma, 2025, Thm. 2]. The constant mode of the advection–diffusion generator has $\lambda_{\min}(A_L) = 0$, which is the worst case for an approximate kernel; this is why $c > 0$ is needed (with $c = 0$ the error at $\lambda = 0$ is $1 - \mathrm{erfc}(1/2\gamma) = O(1/\gamma)$).

3.2 Exact absorption of the analytic-extension phase

Proposition 1 (phase absorption)**.** *Let $A = A_L + \mathrm{i}A_H$ be time independent, $c \in \mathbb{R}$, $t > 0$, and $U_c(t;k) = e^{-\mathrm{i}(A_H + k(A_L + \frac{c}{t}I))t}$. Then $e^{-\mathrm{i}ck}U(t;k) = U_c(t;k)$ for every k. Consequently, for the kernel (5) and the FCC sum (4),*

$$\sum_j w_j V_j = e^c \sum_j w_j^{(0)}\, e^{-\mathrm{i}(A_H + \sin\theta_j\, B_m^{(c)})t}, \qquad B_m^{(c)} = k_{\max}\big(A_L + \tfrac{c}{t}I\big),$$
$$w_j^{(0)} = k_{\max}\cos\theta_j\,\Delta\theta\, g_0(k_j), \qquad g_0(k) = \frac{e^{-(1+k^2)/(4\gamma^2)}}{\pi(1+k^2)}. \tag{6}$$

The weights $w_j^{(0)}$ are real, even in $j \mapsto N_k - 1 - j$, and positive.

Proof. I commutes with A_L and A_H, so $e^{-\mathrm{i}(A_H + kA_L)t}e^{-\mathrm{i}ckI} = e^{-\mathrm{i}(A_H + kA_L + kcI/t)t}$; insert $k = k_j = k_{\max}\sin\theta_j$. □

What changes in the circuit. In the block-encoding of B_m (an LCU over the three bands of A_L), only the rotation that loads the diagonal changes, $\cos(\vartheta_c/2) = 4k_{\max}(d + c/t)/\eta$ with $d = 2D/\Delta x^2$ the diagonal of A_L; the common normalisation must satisfy $\eta \geq 4k_{\max}(d + c/t)$, i.e. the selector length grows by the factor $1 + c/(td)$ (for a general A_L: $\|A_L + \frac{c}{t}I\|/\|A_L\|$). No register, gate or control is added. The output carries the known factor e^{-c}, removed in classical post-processing; the LCU one-norm is $e^c\sum_j w_j^{(0)}$ in both forms, so the success probability is the same as with complex weights.

Remark 1. *(i) On the uniform grid $k_j = hj$ used by LS, $e^{-\mathrm{i}ch_j}$ factorises into single-qubit phase gates [Low and Somma, 2025, Lemma 19]; on the FCC grid it does not, but by Proposition 1 it needs no gates at all. (ii) For the general LS kernel $\hat{f}_{j,y} \propto e^{c(1-\mathrm{i}k)}e^{-(k^2+1)/4\gamma^2}/[(1-\mathrm{i}k)(y+\mathrm{i}k)^{j-1}]$ the same shift removes $e^{-\mathrm{i}ck}$, but $(1-\mathrm{i}k)(y+\mathrm{i}k)^{j-1}$ is real and even on $\mathbb{R}$ only for $(j,y) = (2,1)$; every other member keeps complex weights. (iii) The identity shift itself is standard—it is how non-positive A_L are handled in An et al. [2023], Novikau and Joseph [2025b] and it appears in LS's Sec. 1.1.4—; the point here is that in the FCC circuit it converts the analytic-extension kernel into a real-weight LCU at zero structural cost. (iv) For time-dependent A replace $(c/t)I$ by $l(s)I$ with $\int_0^t l = c$.*

3.3 Admissibility, scaling and the success-probability constraint

Kernel (5) satisfies the approximate-LCHS conditions of LS with $y_0 = 2c\gamma^2$: in the strip $\mathrm{Im}\, z \in [-y_0, 0]$ its only pole is $z = -\mathrm{i}$ (the pole at $+\mathrm{i}$ lies outside), with residue $\sqrt{2/\pi}\, e^0/(-2\mathrm{i}) = \mathrm{i}/\sqrt{2\pi}$ for the $\hat{f}$ normalisation, and $|\hat{f}_2(x - \mathrm{i}y)| \to 0$ uniformly. We verified these conditions numerically for every $(\gamma, c, k_{\max})$ used below (residue error $\leq 6 \times 10^{-17}$; `logs/kernel_derivation/admissibility.txt`); the bound (3), with its contour integral evaluated numerically at $y_0 = 2c\gamma^2$, exceeds the emulator-measured error in all 63 runs with this kernel, by factors 1.8–16. Because the selector length is $\tau = 2\eta t$ with $\eta \propto k_{\max}(\|A_L\| + c/t) + \|A_H\|$, the substituted circuit makes

$$Q_A = O\big(\|At\|\log(1/\epsilon_{\mathrm{LCHS}})/c + \log(1/\epsilon_{\mathrm{LCHS}})\big) \quad \text{versus} \quad Q_A = O\big(\|At\|\log^{1/\beta}(1/\epsilon_{\mathrm{LCHS}})\big) \tag{7}$$

for the ACL kernel; STMC gates scale the same way because the selector dominates. The price is the one-norm: the success probability is $\propto \|\psi(t)\|^2/\alpha^2$. We therefore require *success-probability parity*

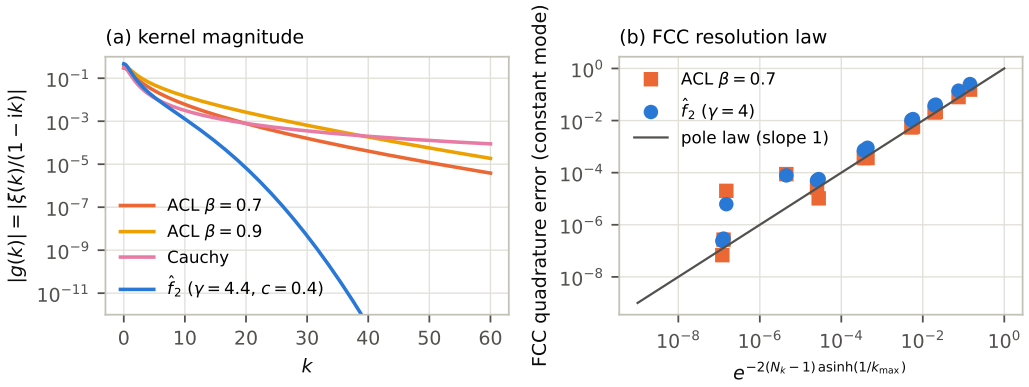


Figure 1: (a) Kernel magnitudes $|g(k)|$: ACL ($\beta = 0.7$ reference, $\beta = 0.9$), Cauchy, and $\hat{f}_2$ with the headline parameters. (b) FCC quadrature error of the constant mode, $|\sum_j w_j - \int_{-k_{\max}}^{k_{\max}} g|$, versus the pole law $e^{-2(N_k-1)\,\mathrm{asinh}(1/k_{\max})}$ for $n_k \in \{5, \ldots, 9\}$, $k_{\max} \in \{5, \ldots, 32\}$ (points below 10^{-9} on the law axis omitted; at the smallest $k_{\max}$ the error saturates at the algebraic endpoint floor). Data: `logs/kernel_derivation/admissibility.json`.

with the reference circuit: the one-norm of the FCC weights, $\alpha_{\hat{f}_2} = e^c \sum_j w_j^{(0)}$ ($\to e^c \operatorname{erfc}(1/2\gamma)$ for large $k_{\max}$), must not exceed the reference's $\alpha_\beta = \sum_j |w_j|$ at its own $k_{\max}$. Asymptotically this forces $c \le \ln \alpha_{0.7} = 0.27$; at the moderate $\gamma \approx 1.3$–6 and small radii selected for $\epsilon_{\mathrm{LCHS}} \ge 10^{-4}$, $\operatorname{erfc}(1/2\gamma) < 1$ and the truncation allow c up to 0.8.

3.4 A resolution law for FCC-based LCHS

In the angle variable the FCC sum is one half of the periodic trapezoidal rule with $2(N_k - 1)$ points per period applied to $G(\theta) = k_{\max} |\cos\theta|\, g(k_{\max} \sin\theta)\, e^{-\mathrm{i} k_{\max} \sin\theta\, x}$, where $x = \lambda t$ for an eigenvalue λ of A_L (for non-commuting A_L, A_H the factor $\|U(t;k)\| \le e^{|\mathrm{Im}\, k|\, \|A_L\| t}$ plays the same role). Every LCHS kernel has a pole at $k = -\mathrm{i}$ (it carries the residue); $\hat{f}_2$ has a second pole at $+\mathrm{i}$ and ξ_β a branch point there. In θ these singularities sit at distance $\mathrm{asinh}(1/k_{\max})$ from the real axis, so, as long as the kernel is negligible at $\pm k_{\max}$ (otherwise the kink of $|\cos\theta|$ at $\pm\pi/2$ adds the algebraic floor analysed in Novikau and Joseph [2025b]), the standard estimate for analytic periodic integrands [Trefethen and Weideman, 2014] gives

$$|\text{FCC error}| \lesssim C\, e^{x}\, e^{-2(N_k-1)\,\mathrm{asinh}(1/k_{\max})} \;\Rightarrow\; k_{\max} \lesssim k^\star_{\max}(n_k, \epsilon_{\mathrm{LCHS}}) \approx \frac{2(N_k - 1)}{\ln(C/\epsilon_{\mathrm{LCHS}})}. \quad (8)$$

Figure 1(b) compares (8) with the measured quadrature error of $\sum_j w_j$ for both kernels. At the 13 points that are not at the algebraic floor, the ratio of measured error to the law with $C = 1$ has median 1.03 (range 0.37–2.09) for the ACL kernel and 2.0 (range 1.75–2.28) for $\hat{f}_2$, whose pole at $+\mathrm{i}$ has the same strength as the one at $-\mathrm{i}$. The consequence is a feasibility condition: a kernel family reaches ϵ_{LCHS} at resolution n_k only if its truncation radius satisfies $R(\epsilon_{\mathrm{LCHS}}) \lesssim k^\star_{\max}(n_k, \epsilon_{\mathrm{LCHS}})$. At small n_k this excludes slowly decaying kernels long before their asymptotic scaling matters—the mechanism behind the kernel-family map in Sec. 5.5.

3.5 Weight preparation by one even QSVT with exact amplitude amplification

With $x = \sin\theta \in [-1, 1]$ the weights (6) sample the even function $w(x) = k_{\max}\sqrt{1 - x^2}\, \Delta\theta\, g_0(k_{\max} x)$. The SIN gate is, for each j, the reflection $\left(\begin{smallmatrix} x_j & \sqrt{1-x_j^2} \\ \sqrt{1-x_j^2} & -x_j \end{smallmatrix}\right)$ on one ancilla, i.e. a block-encoding of $\mathrm{diag}_j(\sin\theta_j)$, so a QSVT with $d+1$ phases on it applies $\mathrm{diag}_j P(\sin\theta_j)$ for any even polynomial $|P| \le 1$. We prepare

$$\mathrm{PREP}\,|0\rangle = \tfrac{1}{\sqrt{N_k}} \sum_j P(x_j)\, |j\rangle |0\rangle_{\mathrm{flags}} + |\perp\rangle, \qquad P(x) \approx \frac{\sqrt{w(x)}}{\sigma}, \quad \sigma = \frac{\sqrt{\alpha_0 / N_k}}{\sin \frac{\pi}{4N_{\mathrm{AA}}+2}}, \quad (9)$$

with $\alpha_0 = \sum_j w_j^{(0)}$ and the smallest N_{AA} for which $\max|P| \le 0.95$. The scale σ makes N_{AA} standard AA rounds map the success amplitude to exactly one [Brassard et al., 2002], so the amplified preparation has no flagged remainder except a term of second order in the polynomial error, $\cos^2((2N_{\mathrm{AA}}+1)\tilde{\theta})$ for the realised angle $\tilde{\theta}$; the left and right preparations (identical, since the weights are real) can then share their flag qubits $a_{\mathrm{qsvt}}, a_{\mathrm{sin}}$, at the price of that flagged branch entering the output (at most 4.0×10^{-6} relative to the output in all our runs, and included in every measured error). Together with the AA marker this gives three weight ancillas—the count of the brute-force oracle emulated in Novikau and Joseph [2025b], which needs two flag qubits because its AA is not exact. The Chebyshev coefficients of $\sqrt{w}/\sigma$ are computed by interpolation and truncated at the smallest even degree d for which the LCHS error obtained with the *realised* node amplitudes stays within 2% of the exact-weight error (and below the paired bound of Sec. 4); the singularities of $\sqrt{w}$ at $x = \pm\mathrm{i}/k_{\max}$ make $d \propto k_{\max}\ln(1/\delta)$, so the smaller $k_{\max}$ of $\hat{f}_2$ also shortens the QSVT (degree 58 at the headline, versus two polynomials of degrees 162 and 163 for the ACL kernel). Phases are computed by QuCF's angle finder (L-BFGS on the symmetric-phase residual of Dong et al. [2021]) with stopping tolerance 10^{-12} and checked three ways: QuCF's own reconstruction (worst deviation 1.3e-12 over all 221 angle sets used), an independent simulator of QuCF's QSVT circuit (worst deviation 1.8e-12), and emulation of the preparation alone (Sec. 5.1). QuCF's QSVT circuit returns $\pm P$ for even and $\pm\mathrm{i}P$ for odd degree; signs are fixed automatically by simulation.

Complex kernels (ACL, $\hat{f}_{j,y}$; needed for the characterisation) use $\sqrt{w} = P_R + \mathrm{i}P_I$ on the continuous branch $\sqrt{|w|}\,e^{\mathrm{i}\arg g/2}$ (the principal branch jumps where $\arg g$ winds past $\pm\pi$; e.g. $\arg g_\beta = \arctan k - \mathrm{Im}(1+\mathrm{i}k)^\beta$), with P_R even and P_I odd since $g(-k) = \overline{g(k)}$, combined by an LCU $(\mathrm{QSVT}_R + \mathrm{QSVT}_I)/2$ on one extra ancilla (the odd QSVT already supplies the factor i) and a Z on that ancilla for the conjugate left preparation: four weight ancillas.

4 Experimental setup

Benchmark. The one-dimensional periodic advection–diffusion equation $\partial_t\psi = -v\partial_x\psi + D\partial_x^2\psi$ with $v = 1$, $D = 0.01$, $x \in [0,1)$, $N_x = 2^{n_x}$ points, $\Delta x = 1/(N_x-1)$, central differences, and a Gaussian initial state centred at 0.5 with width 0.05, exactly as in Novikau and Joseph [2025b]; $n_x = 6$ throughout ($\|A\| = \|A_L\| = 158.76$, $\|A_H\| = 63$). The reference solution $e^{-At}\psi(0)$ is computed at run time with `scipy.linalg.expm`. The error metric is the relative state error $\|\psi_{\mathrm{circ}} - \psi_{\mathrm{exact}}\|/\|\psi_{\mathrm{exact}}\|$, where ψ_{circ} is the emulator's all-ancillas-zero output divided by the circuit's *nominal* normalisation (initial-state AA gain $\times$ weight AA gain / LCU one-norm, times e^{-c} for the shifted kernel), all fixed by circuit parameters and nominal weights—the exact solution is used only to measure the error.

Emulator and circuits. All circuits are executed with QuCF [Novikau, 2024] (QuEST backend [Jones et al., 2019], double precision) on single NVIDIA H200 GPUs; no instance needs more than one GPU (at most 2^{27} amplitudes). We use a copy of the authors' QuCF-OPT-LCHS sources with two additions: weight profiles for the new kernels on the brute-force path and counters for the selector's block-encoding calls (`codebase_notes.md`; diffs in `logs/qucf_patch_*.diff`); the modified binary reproduces the published output bit-for-bit. The circuits are generated from the reference oracle files: initial-state preparation with two AA rounds, U_{A_H}, U_{B_m} and SIN as in Novikau and Joseph [2025b], the selector as QSP of their LCU. Only numerical rotation angles depend on $(k_{\max}, c, \eta)$; the formulas reproduce the published constants to all printed digits.

Selector time. QSP phases for $e^{-\mathrm{i}\tau x}$ were generated by the authors with an external implementation of Haah's method [Haah, 2019] that is not part of the provided code; we therefore reuse the 26 phase sets shipped with QuCF-OPT-LCHS ($1 \le \tau \le 6222.5$), each re-validated by an independent simulation of QuCF's QSP circuit to $\le 4\times10^{-12}$ (`logs/qsp_segments_validation.txt`). For every run, *for every kernel alike*, we set $\eta_{\mathrm{conv}} = 4\sqrt{3/2}\,k_{\max}(2D/\Delta x^2 + c/t)$ (the reference convention; $c = 0$ for exact kernels), choose the product of segments with the fewest phases such that $\sum\tau_s \ge 2\eta_{\mathrm{conv}}t$, and use $\eta = \sum\tau_s/(2t)$, so that the simulated time is exactly t. (The published runs used $\tau = 5\times2489.0$ with $2\eta t = 12447.5$, i.e. they simulate $t = 0.79984$; Table 2.)

Configurations. All configurations, kernels and selection rules were fixed in a pre-registration file before any validation emulation, and every hyper-parameter was frozen by script before the corresponding runs (`code/kernel_plan.md`, `logs/selection/*.json`).

- **Headline** $(n_x, n_k, t, \epsilon_{\mathrm{LCHS}}) = (6, 7, 0.1, 10^{-3})$: $n_k = 7$ is the smallest k-register at which the reference kernel reaches 10^{-3} for this problem (the smallest error its circuit model reaches at $n_k = 6$, over all $k_{\max}$, is 3.6×10^{-3}).
- **Precision sweep** at $(6, 8, 0.1)$: eleven targets $\epsilon_{\mathrm{LCHS}} = 10^{-1.5}, 10^{-1.75}, \ldots, 10^{-4}$.
- **Smallest-register sweep** at $(6, 6, 0.1)$: the same targets; infeasible ones are reported as such.
- **Time sweep** at $(6, 8)$, $\epsilon_{\mathrm{LCHS}} = 10^{-2.5}$, $t \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$, i.e. $\|A\|t \in [7.9, 127]$.
- **Kernel-family characterisation** at $t = 0.1$ over $n_k \in \{5, 6, 7, 8\}$ and the six targets $\epsilon_{\mathrm{LCHS}} \in \{10^{-1.5}, 10^{-2}, \ldots, 10^{-4}\}$ (half-decade steps); families ACL $\beta = 0.7$ (brute force and QSVT–LCU), ACL $\beta = 0.9$, Cauchy, $\hat{f}_2$ with $\alpha \leq 1.295$, $\hat{f}_2$ with $\alpha \leq 3$, and $\hat{f}_{j,y}$ with LS's numerically optimised (j, y, c) (their Table 3, $\gamma = \infty$, interpolated in $\log \epsilon_{\mathrm{LCHS}}$).
- **Reproduction**: five of the published oracle files ($t = 0.8$) re-run with the unmodified build.

Hyper-parameter rules. On the grids $k_{\max} \in \{2, 2.5, \ldots, 70\}$, $c \in \{0.05, \ldots, 2\}$, $\gamma \in \{0.6, 0.65, \ldots, 20\}$, using the classical model of the circuit (it reproduces emulator outputs to $\leq 4.0 \times 10^{-6}$, and to $\leq 1.6 \times 10^{-11}$ once the flagged branch of Sec. 3.5 is added): the reference circuit takes the smallest $k_{\max}$ whose predicted error is $\leq \epsilon_{\mathrm{LCHS}}$; the substituted kernel (*paired rule*) takes the smallest $k_{\max}$—over all (γ, c)—whose predicted error does not exceed the reference's and whose one-norm does not exceed the reference's α_β at its $k_{\max}$; the QSVT degree follows Sec. 3.5. Characterisation families use the standalone rule (smallest $k_{\max}$ with predicted error $\leq \epsilon_{\mathrm{LCHS}}$, under their own α caps). The same rules choose the reference's $k_{\max}$ and the new kernel's parameters, so neither side is tuned on emulator output.

Deviations from the reference setup, disclosed. (i) Times: the reference study used $t \in [0.4, 2.4]$ with $n_k \geq 7$; at $n_k \leq 8$ the larger times are not resolved by FCC for *any* kernel (Sec. 3.4, the e^x factor), so the precision studies use $t = 0.1$ and the time sweep extends to $t = 0.8$. (ii) Normalisation: η is rounded up to the available QSP segments as described above (identically for all kernels). (iii) The shifted kernel changes the diagonal rotation of U_{B_m} (Proposition 1); no other block-encoding change. (iv) The new QSVT weight path uses exact AA; the reference circuit keeps its brute-force weights and non-exact AA, with the number of rounds chosen as the first-lobe maximiser of the success probability (this equals the published choice at $n_k = 7, 9$ and is one round more at $n_k = 8, 10$). (v) Success-probability parity is enforced through the one-norm ($\alpha \leq \alpha_\beta$) relative to the reference circuit at the same $(n_x, n_k, t, \epsilon_{\mathrm{LCHS}})$ rather than to the value ≈ 0.2 quoted for $t = 0.8$; at $t = 0.1$ the reference circuit itself succeeds with probability 0.43–0.50.

5 Results

5.1 Reproduction and verification

Re-running five published oracle files with the unmodified build reproduces the published outputs to machine precision in 5 of 5 cases (bitwise in 3; largest amplitude difference 5.4×10^{-16}, from floating-point reduction order on a different GPU), with identical STMC counts in 5 of 5 (Table 2). Measured against the exact solution, the $(n_k, k_{\max}) = (7, 20)$ run at $t = 0.8$ has relative error 2.12×10^{-2}; this value is set by FCC quadrature at $n_k = 7$ and agrees with our classical model of the same circuit, and the published phase sets simulate $t_{\mathrm{eff}} = 0.79984$ rather than 0.8 (an 8×10^{-4} effect, Sec. 4). Before the sweeps we verified every weight construction on the emulator (Table 3): brute-force weights reproduce their targets to 10^{-14}; exact AA of the QSVT preparations yields success amplitude one; the shifted-real and the unshifted complex $\hat{f}_2$ circuits give the same error (4.797×10^{-3}, relative difference 9.7×10^{-11}), which is Proposition 1 on the emulator; and over all 174 sweep runs the emulator output agrees with an exact classical model of the same circuit (realised QSVT polynomials, realised amplitude amplification including the flagged branch of the shared-flag preparation) to 1.6×10^{-11} (median 4.9×10^{-12}; `logs/emulator_model_check.json`).

Table 1: Main results (emulator-measured). Test: advection–diffusion equation (ADE). Configuration $(n_x, n_k, t, \epsilon_{\mathrm{LCHS}})$ with ϵ_{LCHS} the target. STMC: total single-target multi-controlled gates of the full circuit; Q_A: controlled block-encoding calls of the selector; ancillas include the k-register. Rows grouped by configuration; the first group is the headline. "ref.": reference circuit of [Novikau and Joseph, 2025b] (ACL kernel, $\beta = 0.7$, brute-force weights); "ours": shifted $\hat{f}_2$ kernel, one-QSVT weights, paired hyper-parameters; QSVT–LCU: the reference kernel through the reworked weight path. Full sweep: Table 4.

Method	Test	$(n_x, n_k, t, \epsilon_{\mathrm{LCHS}})$	Final rel. error	STMC	Q_A	Ancillae	Success prob.
ACL $\beta = 0.7$, brute force (ref.)	ADE	$(6, 7, 0.1, 10^{-3})$	9.15×10^{-4}	529,927	10,404	19	0.436
$\hat{f}_2$ shifted, QSVT (ours)	ADE	$(6, 7, 0.1, 10^{-3})$	8.98×10^{-4}	284,837	5,436	19	0.438
ACL $\beta = 0.7$, QSVT–LCU	ADE	$(6, 7, 0.1, 10^{-3})$	9.33×10^{-4}	659,186	10,404	20	0.438
ACL $\beta = 0.7$, brute force (ref.)	ADE	$(6, 6, 0.1, 10^{-1.5})$	2.74×10^{-2}	170,078	3,396	18	0.502
$\hat{f}_2$ shifted, QSVT (ours)	ADE	$(6, 6, 0.1, 10^{-1.5})$	2.55×10^{-2}	61,956	1,228	18	0.502
ACL $\beta = 0.7$, QSVT–LCU	ADE	$(6, 6, 0.1, 10^{-1.5})$	2.79×10^{-2}	175,411	3,396	19	0.504
ACL $\beta = 0.7$, brute force (ref.)	ADE	$(6, 6, 0.1, 10^{-2})$	9.82×10^{-3}	237,799	4,764	18	0.457
$\hat{f}_2$ shifted, QSVT (ours)	ADE	$(6, 6, 0.1, 10^{-2})$	8.59×10^{-3}	90,331	1,788	18	0.482
ACL $\beta = 0.7$, QSVT–LCU	ADE	$(6, 6, 0.1, 10^{-2})$	9.87×10^{-3}	248,546	4,764	19	0.457
ACL $\beta = 0.7$, brute force (ref.)	ADE	$(6, 6, 0.1, 10^{-2.25})$	5.37×10^{-3}	336,403	6,756	18	0.442
$\hat{f}_2$ shifted, QSVT (ours)	ADE	$(6, 6, 0.1, 10^{-2.25})$	4.89×10^{-3}	105,626	2,088	18	0.474
ACL $\beta = 0.7$, QSVT–LCU	ADE	$(6, 6, 0.1, 10^{-2.25})$	5.34×10^{-3}	357,404	6,756	19	0.442
ACL $\beta = 0.7$, brute force (ref.)	ADE	$(6, 6, 0.1, 10^{-2.5})$	infeasible at this n_k (no $k_{\max} \le 70$ reaches ϵ)				
$\hat{f}_2$ shifted, QSVT (ours)	ADE	$(6, 6, 0.1, 10^{-2.5})$	3.06×10^{-3}	139,968	2,764	18	0.466
ACL $\beta = 0.7$, brute force (ref.)	ADE	$(6, 6, 0.1, 10^{-2.75})$	infeasible at this n_k (no $k_{\max} \le 70$ reaches ϵ)				
$\hat{f}_2$ shifted, QSVT (ours)	ADE	$(6, 6, 0.1, 10^{-2.75})$	1.67×10^{-3}	202,839	3,968	18	0.538
ACL $\beta = 0.7$, brute force (ref.)	ADE	$(6, 8, 0.1, 10^{-1.5})$	2.74×10^{-2}	187,546	3,396	20	0.503
$\hat{f}_2$ shifted, QSVT (ours)	ADE	$(6, 8, 0.1, 10^{-1.5})$	2.56×10^{-2}	64,552	1,228	20	0.502
ACL $\beta = 0.7$, QSVT–LCU	ADE	$(6, 8, 0.1, 10^{-1.5})$	2.79×10^{-2}	183,427	3,396	21	0.504
ACL $\beta = 0.7$, brute force (ref.)	ADE	$(6, 8, 0.1, 10^{-2})$	9.79×10^{-3}	258,003	4,764	20	0.455
$\hat{f}_2$ shifted, QSVT (ours)	ADE	$(6, 8, 0.1, 10^{-2})$	8.63×10^{-3}	94,167	1,788	20	0.482
ACL $\beta = 0.7$, QSVT–LCU	ADE	$(6, 8, 0.1, 10^{-2})$	9.85×10^{-3}	260,274	4,764	21	0.457
ACL $\beta = 0.7$, brute force (ref.)	ADE	$(6, 8, 0.1, 10^{-2.5})$	2.68×10^{-3}	384,688	7,224	20	0.438
$\hat{f}_2$ shifted, QSVT (ours)	ADE	$(6, 8, 0.1, 10^{-2.5})$	2.62×10^{-3}	160,318	3,032	20	0.486
ACL $\beta = 0.7$, QSVT–LCU	ADE	$(6, 8, 0.1, 10^{-2.5})$	2.73×10^{-3}	407,181	7,224	21	0.442
ACL $\beta = 0.7$, brute force (ref.)	ADE	$(6, 8, 0.1, 10^{-3})$	9.74×10^{-4}	548,473	10,404	20	0.434
$\hat{f}_2$ shifted, QSVT (ours)	ADE	$(6, 8, 0.1, 10^{-3})$	8.93×10^{-4}	291,463	5,436	20	0.438
ACL $\beta = 0.7$, QSVT–LCU	ADE	$(6, 8, 0.1, 10^{-3})$	9.92×10^{-4}	622,330	10,404	21	0.438
ACL $\beta = 0.7$, brute force (ref.)	ADE	$(6, 8, 0.1, 10^{-3.5})$	2.94×10^{-4}	650,596	12,408	20	0.432
$\hat{f}_2$ shifted, QSVT (ours)	ADE	$(6, 8, 0.1, 10^{-3.5})$	2.73×10^{-4}	377,874	6,952	20	0.453
ACL $\beta = 0.7$, QSVT–LCU	ADE	$(6, 8, 0.1, 10^{-3.5})$	2.98×10^{-4}	767,271	12,408	21	0.437
ACL $\beta = 0.7$, brute force (ref.)	ADE	$(6, 8, 0.1, 10^{-4})$	9.77×10^{-5}	834,142	15,972	20	0.432
$\hat{f}_2$ shifted, QSVT (ours)	ADE	$(6, 8, 0.1, 10^{-4})$	9.60×10^{-5}	433,217	7,972	20	0.451
ACL $\beta = 0.7$, QSVT–LCU	ADE	$(6, 8, 0.1, 10^{-4})$	9.90×10^{-5}	1,005,775	15,972	21	0.436

5.2 Headline configuration

At $(n_x, n_k, t, \epsilon_{\mathrm{LCHS}}) = (6, 7, 0.1, 10^{-3})$ the reference circuit needs $k_{\max} = 29$ and 529,927 STMC gates (error 9.15×10^{-4}, success probability 0.436). The shifted $\hat{f}_2$ kernel with $(\gamma, c) = (4.4, 0.4)$ needs $k_{\max} = 14.5$ and 284,837 gates—$1.86\times$ fewer—with a smaller error (8.98×10^{-4}), a higher success probability (0.438) and the same 19 ancillas (Table 1, Fig. 2). The selector accounts for 525,427 and 274,523 gates respectively and Q_A falls by $1.91\times$; outside the selector (both weight preparations with their AA, plus the common initial-state preparation) the circuit has 10,314 gates with the one-QSVT weights (degree 58) against 4,500 with the brute-force weights. Running the reference kernel through the reworked path (QSVT–LCU, degrees 162 and 163) gives the same accuracy (9.33×10^{-4}) with 133,759 gates outside the selector and one more ancilla: real weights matter for the weight preparation, but the dominant saving comes from the smaller truncation radius.

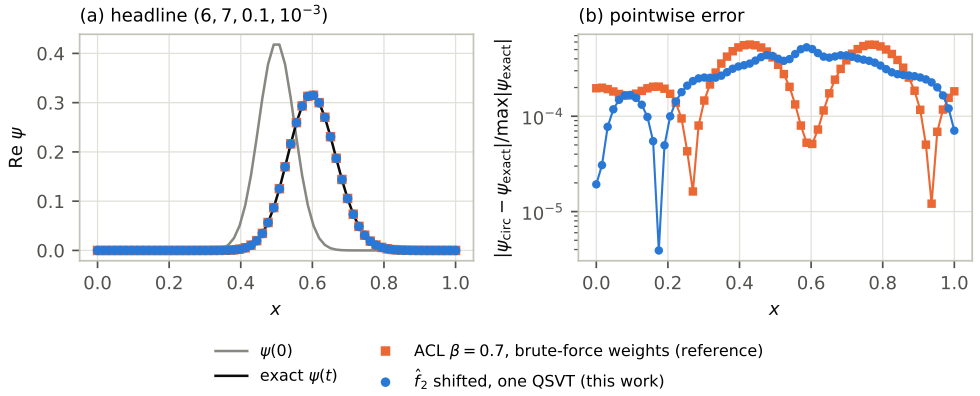


Figure 2: Headline configuration. (a) Initial state, exact solution at $t = 0.1$, and emulated outputs of the reference and of the substituted circuit (post-processed with the nominal normalisation). (b) Pointwise error relative to $\max|\psi_{\text{exact}}|$.

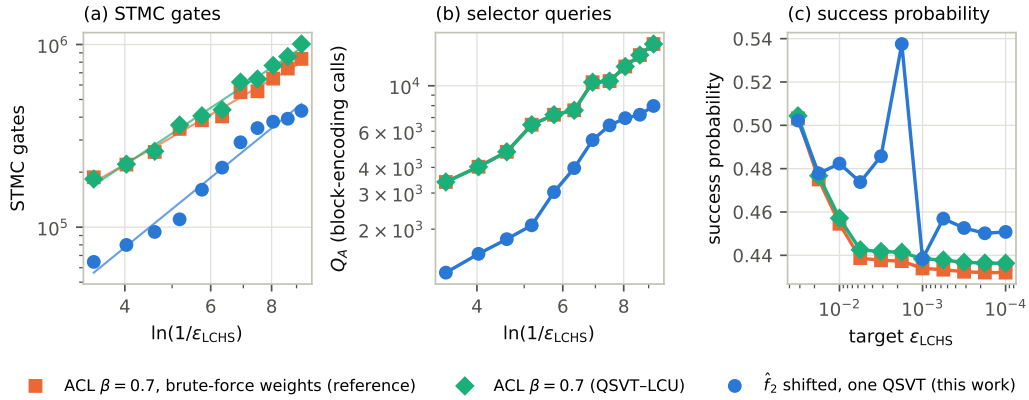


Figure 3: Precision sweep at $(n_x, n_k, t) = (6, 8, 0.1)$, eleven targets $\epsilon_{\text{LCHS}} \in [10^{-4}, 10^{-1.5}]$, paired hyper-parameters. (a) STMC gates, (b) selector block-encoding calls Q_A (log–log axes in $\ln(1/\epsilon_{\text{LCHS}})$), (c) success probability. Every point is one emulation; numbers in Table 4.

5.3 Precision dependence at fixed resolution

At $(6, 8, 0.1)$ the paired rule meets every one of the eleven targets for both kernels. The substituted kernel needs fewer STMC gates at all 11 targets, with gate ratios 1.59–3.13 and Q_A ratios 1.64–3.09 (Fig. 3, Table 4); the ratio is largest at low precision. Its error is $\leq$ the reference's at 11 and its success probability $\geq$ the reference's at 10 of the 11 targets. The exception, 0.5022 against 0.5032 at $10^{-1.5}$, comes from the reference's own error: its 2.7×10^{-2} approximation error makes its output norm 1.6% too large (ours: 0.5%). At the exact solution's norm the two circuits would succeed with probability 0.497 and 0.488; this nominal success probability, which the one-norm rule controls, is higher for the substituted kernel at all 11 targets. Pure power-law fits $N_{\text{gates}} \approx a \ln^p(1/\epsilon_{\text{LCHS}})$ over the window give $(a, p) = (26{,}053,\ 1.54 \pm 0.06)$ for the reference and $(3{,}842,\ 2.17 \pm 0.12)$ for the substituted kernel—a much smaller prefactor with a larger exponent (Q_A exponents: 1.60 ± 0.06 and 2.13 ± 0.12); restricted to $\epsilon_{\text{LCHS}} \leq 10^{-3}$ the fits are 1.55 ± 0.22 and 1.28 ± 0.17. The fitted exponent of the optimal-class kernel is therefore *not* smaller in this window; Sec. 6.2 explains why and what it implies.

5.4 Dependence on $\|At\|$

At fixed target $\epsilon_{\text{LCHS}} = 10^{-2.5}$ and $n_k = 8$ (Fig. 4, Table 5), the gate counts grow close to linearly in $\|A\|t$ over $t \in [0.05, 0.8]$ ($\|A\|t \in [7.9, 127]$): $N_{\text{gates}} \approx a(\|A\|t)^p$ with $(a, p) =$

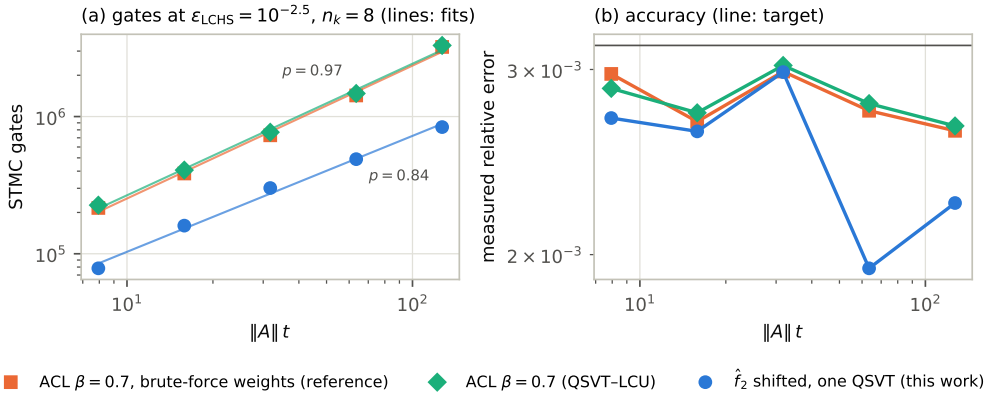


Figure 4: Time sweep at $(n_x, n_k) = (6, 8)$, target $\epsilon_{\mathrm{LCHS}} = 10^{-2.5}$. (a) STMC gates versus $\|A\|t$ (log–log) with power-law fits and their exponents p. (b) Measured error.

$(27{,}178,\ 0.97 \pm 0.04)$ for the reference and $(14{,}766,\ 0.84 \pm 0.04)$ for the substituted kernel (prefactor ratio 1.84, exponent difference -0.12 ± 0.05). The substituted circuit needs 2.40–3.84× fewer gates at the five times, with error $\leq$ and success probability $\geq$ the reference's at 5 of 5. Its smaller exponent has two sources: the t-independent part of its selector, the shift c/t of Proposition 1 (worth 11% of η at $t = 0.05$ and 0.5% at $t = 0.8$), and a slightly smaller $k_{\max}$ at the longest times. The reduction is thus a prefactor at an exponent in $\|At\|$ that is not worse than the reference's.

5.5 Smallest k-register and kernel-family map

With a 6-qubit k-register the reference kernel reaches 4 of the eleven targets (down to $10^{-2.25}$) and the substituted kernel 6 under the same success-probability cap (down to $10^{-2.75}$), with 2.63–3.18× fewer gates where both are feasible (Table 1). Lifting the cap to $\alpha \leq 3$ reaches 10^{-4} at $n_k = 6$ (success probability 0.083). This is the resolution law (8) at work: at $n_k = 6$ the reference kernel needs $k_{\max} = 19$ already for $10^{-2.25}$, where the pole term $e^{-126\,\mathrm{asinh}(1/k_{\max})} \approx 1.3 \times 10^{-3}$ is within a factor of a few of the target, so the larger radii needed for smaller ϵ_{LCHS} are limited by quadrature error; the substituted kernel needs $k_{\max} = 10$ for $10^{-2.75}$, where the pole term is 3.4×10^{-6}.

Figure 5 and Tables 6–7 map all families over $n_k \in \{5, \ldots, 8\}$ and six targets (22 of 24 cells reachable by at least one family; the reference kernel reaches 14). Three rankings answer the question "which kernel is best where":

- *At matched success probability* (one-norm $\leq \alpha_{0.7}$: the ACL $\beta = 0.7$ circuits, Cauchy, and the parity-capped $\hat{f}_2$), the parity-capped $\hat{f}_2$ is the cheapest circuit in 18 of the 18 cells this class can reach, at every n_k; below $\approx 10^{-3}$ at $n_k \leq 6$ no member of the class is feasible.
- *Per amplitude-amplified success* ($N_{\mathrm{gates}}/\sqrt{P_{\mathrm{succ}}}$, all families), the parity-capped $\hat{f}_2$ is cheapest in 12 cells—all targets $\epsilon_{\mathrm{LCHS}} \geq 10^{-2.5}$—and the rational $\hat{f}_{j,y}$ with Low and Somma's optimised parameters in the remaining 10 cells, $\epsilon_{\mathrm{LCHS}} \leq 10^{-3}$ (by 1.35× at $(n_k, \epsilon_{\mathrm{LCHS}}) = (8, 10^{-4})$), although it needs complex weights and runs at success probability 0.17–0.33 (parity-capped $\hat{f}_2$: 0.44–0.50).
- *By raw gate count*, ignoring success probability, the uncapped $\hat{f}_2$ ($\alpha \leq 3$, $P_{\mathrm{succ}} = 0.08$–0.27) or $\hat{f}_{j,y}$ is cheapest everywhere.

The exact kernels—ACL with either β, and Cauchy—are never the cheapest (0 and 0 cells in the last two rankings). Small registers or $\epsilon_{\mathrm{LCHS}} \leq 10^{-3}$ admit only the Gaussian-damped or rational optimal-class kernels, as predicted by the resolution law; ACL $\beta = 0.9$ pays $\alpha_{0.9} = 2.0$ without a smaller radius, and Cauchy has the best success probability ($\alpha = 1$) but the largest radius.

6 Discussion

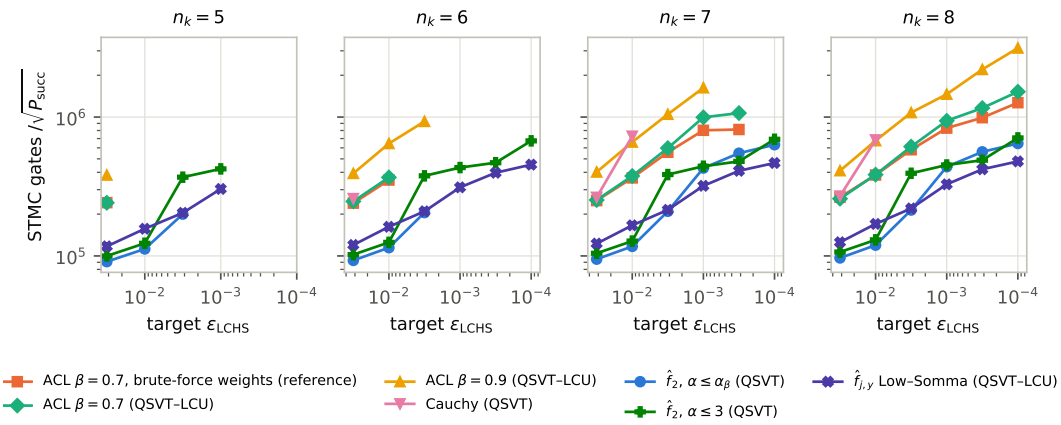


Figure 5: Kernel-family characterisation at $t = 0.1$: amplitude-amplified cost $N_{\text{gates}}/\sqrt{P_{\text{succ}}}$ of the cheapest emulated circuit meeting each target, per k-register size. Missing points: target not reachable by that family at that n_k. Raw gates and success probabilities in Tables 6–7.

6.1 Mechanism: why $\hat{f}_2$ is the right kernel for this circuit at these resolutions

The selector is everything. In the reference and in the substituted circuits at least 93.3% of the STMC gates sit in the selector (the QSVT–LCU preparation of the complex reference weights takes up to 20%), whose length is $2\eta t$ with $\eta \propto k_{\max}$; at matched success probability the gate ratio is therefore essentially the ratio of truncation radii, corrected by $1 + c/(td)$ for the shift and by the weight preparation. Everything that makes $\hat{f}_2$ win acts through $k_{\max}$.

Gaussian versus stretched-exponential tails. $|\xi_\beta(k)| \sim e^{-0.45|k|^{0.7}}$ for the reference kernel, whereas $\hat{f}_2$ falls as $e^{-k^2/4\gamma^2}/k^2$ (Fig. 1a). Truncation of $\hat{f}_2$ costs $k_{\max} \approx 2c\gamma^2$ with γc growing only like $\sqrt{\ln(1/\epsilon_{\text{LCHS}})}$. The price is an approximation error at the bottom of the spectrum of $A_L t$; for the normal advection–diffusion generator it is the scalar error at the conserved mode, $e^c f_2(c;\gamma) - 1 = O(e^{c-\gamma^2 c^2}/\gamma^3 c^2)$ (headline: 9.8×10^{-4}, measured state error 8.98×10^{-4}; the bound (3) at the same parameters gives 1.8×10^{-3}). The paired rule exploits this: it buys a small $k_{\max}$ with a moderate c and spends the rest of the one-norm budget α_β on γ.

The resolution law favours small radii. By (8), an n_k-qubit register supports $k_{\max} \lesssim 2(N_k - 1)/\ln(C/\epsilon_{\text{LCHS}})$. The reference kernel's radius outgrows this bound before its truncation error is small enough: over all $k_{\max}$ its circuit model reaches at best 3.6×10^{-3} at $n_k = 6$ and 3.0×10^{-4} at $n_k = 7$, which is exactly where its runs disappear from Tables 1 and 6. Under the same one-norm cap the substituted kernel's floors are 1.4×10^{-3} and 8.9×10^{-5}, and with $\alpha \le 3$ they are 5.2×10^{-5} and 8.7×10^{-7}. In this framework the kernel therefore also determines the k-register (and hence the state-vector memory) needed for a given precision.

Real, positive weights are cheap to prepare. After phase absorption the preparation is one even QSVT whose degree is $\propto k_{\max}\ln(1/\delta)$ because of the branch points of $\sqrt{w}$ at $x = \pm i/k_{\max}$; the smaller radius also shortens it (degree 58 vs. two polynomials of degree ≈162 for the reference kernel at the headline), and exact AA removes the second flag qubit. Positivity also means that the LCU has no cancellation: its one-norm is $\alpha_0 e^c$ with $\alpha_0 = \sum_j w_j^{(0)} < \operatorname{erfc}(1/2\gamma) < 1$, which is what allows c as large as 0.7–0.8 at low precision without losing success probability.

Why not the other families. ACL $\beta = 0.9$ has a better exponent ($1/\beta = 1.11$) but $\cos(\beta\pi/2) = 0.16$ and $\alpha = 2.01$: in the window its radius is not smaller than for $\beta = 0.7$ and its success probability is halved. Cauchy has $\alpha = 1$ and the best success probability but an algebraic tail, so it is feasible only at $\epsilon_{\text{LCHS}} \gtrsim 10^{-2}$. The uncapped $\hat{f}_2$ and the rational $\hat{f}_{j,y}$ have the smallest radii—Low and Somma optimise exactly the product αR—but their one-norms ≈ 2–3 cut the success probability to 0.08–0.27 and 0.17–0.33, respectively, and $\hat{f}_{j,y}$ needs complex weights (two QSVTs and an LCU ancilla). Per amplitude-amplified success the rational kernel nevertheless overtakes the parity-capped $\hat{f}_2$ below

$\epsilon_{\mathrm{LCHS}} \approx 10^{-3}$ (Sec. 5.5): its radius grows more slowly once γ of $\hat{f}_2$ must be large. The parity-capped $\hat{f}_2$ is therefore the best choice among the kernels we tested *for the circuit as specified*—one shot, success probability at least that of the reference, no amplitude amplification of the whole LCHS circuit—because it is the only optimal-class kernel here that is simultaneously (i) Gaussian-tailed, giving the smallest radius per unit one-norm at moderate precision, (ii) real-weighted after absorption, giving the cheapest preparation without an extra register, and (iii) tunable to the reference one-norm. If the whole circuit is to be amplitude-amplified anyway, the rational kernel is the better choice at $\epsilon_{\mathrm{LCHS}} \leq 10^{-3}$, and a real-weight rational kernel (Remark (ii) excludes it within the $\hat{f}_{j,y}$ family) is an open target.

6.2 Failure analysis: where the substituted kernel does not help

No visible exponent gain in $\log(1/\epsilon_{\mathrm{LCHS}})$. Over $[10^{-4}, 10^{-1.5}]$ the fitted exponent of the substituted kernel is larger than the reference's (Sec. 5.3). The cause is the success-probability cap: as ϵ_{LCHS} decreases, γ must grow, $\mathrm{erfc}(1/2\gamma) \to 1$, and c must shrink (from 0.7 at $10^{-1.5}$ to 0.35 at 10^{-4}) to keep the one-norm $\approx e^{c}\mathrm{erfc}(1/2\gamma)$ below α_β; since $k_{\max} \approx 2\ln(1/\epsilon_{\mathrm{LCHS}})/c$, the radius grows faster than $\ln(1/\epsilon_{\mathrm{LCHS}})$ until c saturates at $\ln \alpha_\beta \approx 0.27$. Problem-independent designs show the same. Taking, for every ϵ_{LCHS}, the smallest radius that the bound (3) certifies under the same cap (`code/theory_scaling.py`, `logs/theory_scaling.json`) gives window exponents 1.50 for the reference kernel and 1.54 (LS's closed-form estimate of the contour term) or 1.67 (contour integral evaluated numerically) for $\hat{f}_2$; between 10^{-8} and 10^{-12} they are 1.53 versus 1.25 and 1.39, still above the asymptotic value 1. The asymptotic advantage is real but lies far below the precisions that $n_k \leq 8$ can resolve; in the window the gain is a radius prefactor.

Worst-case versus problem-specific parameters. Without a model of the specific problem one would choose parameters from the bound (3). Such worst-case designs keep part of the gain: under the same cap the reference-to-$\hat{f}_2$ radius ratio is 1.46–1.55 in our window with LS's closed-form contour estimate (the basis of their Theorem-2 rule; $R \approx 38$ versus 57 at 10^{-4}) and 2.14–2.52 with the contour integral evaluated numerically ($R \approx 27$). The closed form replaces $|1+z^2|$ on the shifted contour $z = k - \mathrm{i}y_0$ by its real-line value $1+k^2$; since $|1+z^2| \approx y_0^2 = 4c^2\gamma^4$ there, keeping it gives the factor $\approx (2\sqrt{\pi}\gamma^3 c^2)^{-1} \sim 10^{-2}$. The problem-specific selection that we emulate gives $k_{\max}$ ratios 1.8–3.7: 1.18–1.56 times the numerically evaluated bound's ratio at $\epsilon_{\mathrm{LCHS}} \geq 10^{-2.5}$ and 0.81–1.01 times it at higher precision. This tracks how loose the bound is for $\hat{f}_2$ on this problem: it exceeds the measured error by 5.6–8.4× at $\epsilon_{\mathrm{LCHS}} > 10^{-3}$ in the precision sweep but by only 2.0–3.9× at $\epsilon_{\mathrm{LCHS}} \leq 10^{-3}$ (1.8–16× over all 63 $\hat{f}_2$ runs). This problem-specific advantage is not an artefact of the normal generator or of the Gaussian initial state. Repeating the paired selection classically (`code/robustness_check.py`, coarser grids, `logs/robustness.json`) with the operator-norm error—the worst case over initial states—gives $k_{\max}$ ratios 1.6–3.2 at the 6 targets that both kernels reach in operator norm at $n_k = 8$; for a non-normal generator (variable velocity $v = 1 + 0.5\sin 2\pi x$, $\|[A_L, A_H]\| = 232$, A_L shifted to $\lambda_{\min} = 0$) the ratios are 2.2–4.2 for the Gaussian state (11 targets) and 1.8–3.3 in operator norm.

Small registers at high precision cost success probability. Under the parity cap $\hat{f}_2$ reaches at most $10^{-2.75}$ at $n_k = 6$; reaching 10^{-4} there requires $\alpha \approx 3$ and success probability 0.083.

Very short times. The shift adds c/t to the diagonal of A_L; at $t = 0.05$ this makes the selector 13% longer. For $\|A_L\|t \lesssim c$ the shift would dominate; the phase could then instead be put on the weights (complex weights, two QSVTs).

Weight preparation at high precision. The brute-force preparation costs $\propto 2^{n_k}$ rotations independently of ϵ_{LCHS}, the QSVT $\propto d\, n_k$ with $d \propto k_{\max}\ln(1/\delta)$. At $n_k = 8$ the gates outside the selector are fewer with the QSVT weights at 7 of the 11 targets ($\epsilon_{\mathrm{LCHS}} \geq 10^{-3}$) and up to 2.0× more at the highest precision; at the headline ($n_k = 7$) the brute force is cheaper (4,500 vs. 10,314 gates). Either way these gates are at most 6.7% of the total here; the QSVT is the scalable choice for larger registers.

Rejected choices. $c = 0$ (real-even kernel without shift) fails at the conserved mode ($O(1/\gamma)$ error); compactly supported mollifiers of $e^{-|x|}$ (e.g. Kaiser–Bessel or bump functions) give *exact* kernels, which the ACL no-go restricts to sub-exponential decay; a fixed-c rule following LS Theorem 2 gives an affine $k_{\max}(\ln 1/\epsilon_{\mathrm{LCHS}})$ but larger radii than the reference at $\epsilon_{\mathrm{LCHS}} \gtrsim 10^{-2}$; the LS cost-optimal operating point ($\alpha \approx$ 2–3) violates success-probability parity. These alternatives and

the hyper-parameter study that led to the paired rule are documented in `attempts_log.md` and `logs/hyperparam`.

6.3 Classical cost

The substituted kernel adds classical work only in the weight preparation (Table 8): evaluating the N_k FCC weights takes 14 μs (reference kernel: 23 μs), the Chebyshev interpolant 38 ms, and QuCF's angle finder 1.2 ms per run (median; maximum 19 ms) for the real kernels versus 2.3 ms (maximum 869 ms) for the two-polynomial complex preparations. The reference circuit needs no angle finding (its rotation angles are closed-form), and both use the same pre-computed selector phases. Hyper-parameter selection costs 0.09 s for the reference kernel and 5.8 s for the full $(k_{\max}, c, \gamma)$ table of $\hat{f}_2$ at one configuration ($137 \times 40 \times 389$ classical LCHS sums). All of this is negligible next to a single emulation (median 22.8 s for the real-weight QSVT circuits, 90.7 s for the reference circuits, on one H200; the reference circuits are longer) and is a one-off per configuration on hardware. The classical cost does grow with the precision through the QSVT degree ($\propto k_{\max} \ln(1/\delta)$), but for the angle finder used here it remains below a second up to degree 475.

7 Related work

LCHS and Schrödingerisation. LCHS was introduced with the Cauchy kernel [An et al., 2023]; An, Childs and Lin introduced the exact kernel family (2), near-optimal in all parameters, and proved that exact kernels cannot decay exponentially [An et al., 2025a]. Schrödingerisation [Jin et al., 2022, 2025a, Hu et al., 2024] is the same identity seen through a warped-phase dilation; smooth initialisations of the dilated variable reach optimal matrix-query dependence [Jin et al., 2025b]. LCHS has been extended to eigenvalue transformations through the Laplace transform [An et al., 2024] and Weyl calculus [Ni and Ying, 2026], to infinite-dimensional operators [Lu et al., 2025], to commutator-scaling product-formula implementations with sinh–sinh quadrature [Aftab et al., 2026], and to hybrid oscillator–qubit hardware in which the kernel is loaded into a bosonic mode [Das et al., 2026]. Lower bounds and fast-forwarding limits are discussed in An et al. [2025b].

Optimal-class kernels. Approximating exponential decay by entire functions—"analytic extension"—was explored for Fourier-series QSP in de Lima Silva et al. [2022] and with erfc-type approximations in Holmes et al. [2022]. Low and Somma [2025] formalised the approximate-LCHS conditions (3), introduced $\hat{f}_2$ and the family $\hat{f}_{j,y}$, proved $\alpha R = O(\log 1/\epsilon_{\mathrm{LCHS}})$, showed that a uniform trapezoidal rule converges exponentially, and gave a uniform-grid circuit whose weight preparation uses QSVT on a block-encoded number operator plus phase gates for $e^{-\mathrm{i}ckj}$; their numerically optimised kernels are within a factor ≈ 3 of a QSVT lower bound. Huang and An [2025] construct kernels from the Fourier transform alone and report up to $1.8\times$ query and $8.3\times$ depth reductions at $\epsilon_{\mathrm{LCHS}} \le 10^{-8}$; Pocrnic et al. [2025] optimise constant factors of the ACL kernel and its quadrature. All of these are analyses of query complexity or circuit depth on their own discretisations; none of them evaluates an optimal-class kernel inside the FCC/single-QSP circuit.

Explicit circuits and emulation. The sinusoidal map and its single-QSP selector are due to Novikau and Joseph [2025b,a], who also showed that it is equivalent to FCC quadrature and analysed its multiple asymptotic convergence regimes; their emulations use the ACL kernel and brute-force weights. Earlier explicit LCHS/Schrödingerisation circuits used sequences of controlled evolutions [Hu et al., 2024, Sato et al., 2025]. Our resolution law (8) is the angle-variable counterpart of LS's uniform-grid estimate (their step size is limited by the same pole at $k = -\mathrm{i}$); it explains why the FCC circuit's precision at fixed n_k is kernel-limited. QSVT [Gilyén et al., 2019] and its phase finding [Dong et al., 2021, Haah, 2019, Bernard and Wiebe, 2026] are used as standard tools; exact amplitude amplification by a reduced initial amplitude is textbook [Brassard et al., 2002].

Positioning. The kernel $\hat{f}_2$ is Low and Somma's; we claim no new kernel. What is new is (i) the exact absorption of its phase into the FCC block-encoding, which makes it the unique member of their family with real weights in this circuit and enables a one-QSVT preparation without extra registers, (ii) the resolution law that predicts which kernel is usable at which k-register size, and (iii) an end-to-end emulator comparison at matched accuracy and success probability, which shows a 1.59–$3.13\times$ gate reduction in practice but not (yet) the asymptotic exponent.

8 Limitations and conclusions

Limitations. (i) One emulated benchmark: the advection–diffusion generator with periodic boundaries is normal and its conserved mode sits at $\lambda_{\min}(A_L) = 0$. The classical robustness check (Sec. 6.2) indicates that the advantage survives worst-case initial states and a non-normal variable-velocity generator, but we did not emulate a non-normal or Koopman–von Neumann problem. (ii) Small registers only ($n_x = 6$, $n_k \leq 8$, at most 2^{27} amplitudes) and short times; the time sweep reaches $\|A\|t = 127$. (iii) STMC gates are a coarse metric: an STMC gate with many controls costs $O(n_c)$ elementary gates [Barenco et al., 1995], so circuits with the same STMC count but different control widths differ in elementary cost; both circuits here have the same register layout. (iv) The selector is assembled from pre-computed QSP segments; with phases for arbitrary τ both circuits would shrink by a few per cent, identically. (v) Hyper-parameters are selected with the classical model of the same problem instance, for both kernels; a practitioner limited to the problem-independent bound (3) would obtain radius ratios of 1.46–1.55 (LS's closed-form estimate) or 2.14–2.52 (contour integral evaluated numerically) instead of the emulated 1.8–3.7 (Sec. 6.2).

Conclusions. The explicit FCC/single-QSP LCHS circuit can host an optimal-class kernel without any structural change. For Low and Somma's $\hat{f}_2$ the analytic-extension phase is absorbed exactly into the block-encoding that the circuit already contains, leaving real, even and positive weights that one even-parity QSVT prepares with the reference circuit's ancilla budget. On the emulated benchmark this cuts the STMC gate count by $1.86\times$ at the headline configuration and by 1.59–$3.13\times$ across eleven precision targets at matched accuracy and one-norm (hence success probability), with unchanged linear scaling in $\|At\|$, and it extends the precision reachable with a small k-register because, by the FCC resolution law, the reachable precision at fixed n_k is set by the kernel's truncation radius. The asymptotic $\log(1/\epsilon_{\mathrm{LCHS}})$ advantage itself is not yet visible at $\epsilon_{\mathrm{LCHS}} \geq 10^{-4}$: in this window the optimal-class kernel is a better *constant*, not yet a better *exponent*. Emulating non-normal generators and the Koopman–von Neumann problem, and problem-independent parameter rules built on the numerically evaluated approximate-LCHS bound rather than its closed-form estimate, are the natural next steps.

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A Reproduction of the reference emulations

Table 2 compares our re-runs of published oracle files with the published outputs (Sec. 5.1).

Table 2: Re-runs of five published oracle files of Novikau and Joseph [2025b] ($n_x = 6$, $t = 0.8$, ACL $\beta = 0.7$, brute-force weights) with the unmodified QuCF-OPT-LCHS build. $\max|\Delta|$: largest difference between our and the published output amplitudes. Errors are relative state errors against $e^{-At}\psi(0)$ at the nominal $t = 0.8$ and at the time $t_{\text{eff}} = \sum \tau_s/(2\eta)$ that the published QSP phases actually simulate. Source: `runs/results_R.jsonl`.

case	STMC (ours)	STMC (published)	$\max\|\Delta\|$	succ. prob.	error ($t = 0.8$)	t_{eff}	error (t_{eff})
nk7 kmax20	2,692,135	2,692,135	0 (bitwise)	0.215	2.12×10^{-2}	0.79984	2.12×10^{-2}
nk8 kmax20	2,752,429	2,752,429	5.4×10^{-16}	0.214	4.22×10^{-3}	0.79984	4.18×10^{-3}
nk9 kmax20	2,826,923	2,826,923	3.7×10^{-16}	0.216	4.16×10^{-3}	0.79984	4.11×10^{-3}
nk7 kmax40	5,325,230	5,325,230	0 (bitwise)	0.216	1.05×10^{-1}	0.80000	1.05×10^{-1}
nk8 kmax40	5,437,664	5,437,664	0 (bitwise)	0.213	1.87×10^{-2}	0.80000	1.87×10^{-2}

B Verification of the weight constructions

Table 3 lists the verification circuits that were emulated before any sweep (Sec. 5.1).

Table 3: Stand-alone and end-to-end verification circuits (run before the sweeps; $n_x = 6$, $t = 0.1$). "Weights only": the k-register amplitudes a_j after PREP and AA, compared through $a_j^2\alpha_d/G_{\text{AA}}$ with the target weights w_j (maximum relative deviation; for QSVT this is the polynomial approximation error of $\sqrt{w}$, which is concentrated near $\theta = \pm\pi/2$ and does not affect the LCHS error at the chosen degree); success probability equal to one certifies exact AA. "Full LCHS": relative state error and agreement between emulator output and the classical model of the same circuit. Source: `logs/verification/verification_summary.json`.

circuit	kernel	weights	n_k	$k_{\max}$	degree	weight dev. / error	succ. prob.	emulator vs model
weights only	ACL $\beta = 0.7$	brute force	6	12.5	–	1.24×10^{-14}	0.999726	–
weights only	ACL $\beta = 0.7$	QSVT	6	12.5	60	2.05×10^{-3}	1.000000	–
weights only	$\hat{f}_2$ (shifted, real)	brute force	6	5	–	1.45×10^{-14}	0.995653	–
weights only	$\hat{f}_2$ (shifted, real)	QSVT	6	5	20	9.58×10^{-3}	1.000000	–
weights only	$\hat{f}_{j,y}$	QSVT	7	8	40	1.15×10^{-3}	1.000000	–
full LCHS	ACL $\beta = 0.7$	brute force	6	12.5	–	9.82×10^{-3}	0.4566	6.4×10^{-12}
full LCHS	ACL $\beta = 0.7$	QSVT	6	12.5	60	9.81×10^{-3}	0.4568	3.4×10^{-11}
full LCHS	Cauchy	QSVT	6	6	40	7.02×10^{-2}	0.8440	2.1×10^{-9}
full LCHS	$\hat{f}_2$ (shifted, real)	brute force	6	5	–	4.80×10^{-3}	0.4698	3.8×10^{-12}
full LCHS	$\hat{f}_2$ (shifted, real)	QSVT	6	5	20	4.80×10^{-3}	0.4738	6.0×10^{-11}
full LCHS	$\hat{f}_2$ (unshifted, complex)	brute force	6	5	–	4.80×10^{-3}	0.4732	1.5×10^{-12}
full LCHS	$\hat{f}_{j,y}$	QSVT	7	8	40	2.73×10^{-4}	0.2047	6.4×10^{-12}

C Complete precision sweep

Table 4 lists every target of the precision sweep of Sec. 5.3 (Fig. 3).

Table 4: All eleven targets of the precision sweep at $(n_x, n_k, t) = (6, 8, 0.1)$: reference circuit (ACL $\beta = 0.7$, brute-force weights) versus the shifted $\hat{f}_2$ kernel with QSVT weights (paired rule). Last column: STMC-gate ratio reference/new. Source: `runs/results_main.jsonl`, sweep `P8`.

	ACL $\beta = 0.7$ (reference)				$\hat{f}_2$, QSVT (this work)				
ϵ_{LCHS}	$k_{\max}$	error	STMC	succ.	$k_{\max}$	error	STMC	succ.	ratio
$10^{-1.5}$	9	2.74×10^{-2}	187,546	0.503	2.5	2.56×10^{-2}	64,552	0.502	2.91
$10^{-1.75}$	10.5	1.46×10^{-2}	220,099	0.475	3.5	1.14×10^{-2}	79,899	0.478	2.75
10^{-2}	12.5	9.79×10^{-3}	258,003	0.455	4	8.63×10^{-3}	94,167	0.482	2.74
$10^{-2.25}$	18.5	5.18×10^{-3}	345,136	0.439	5	4.82×10^{-3}	110,402	0.474	3.13
$10^{-2.5}$	20.5	2.68×10^{-3}	384,688	0.438	7.5	2.62×10^{-3}	160,318	0.486	2.40
$10^{-2.75}$	21.5	1.75×10^{-3}	403,645	0.437	10	1.66×10^{-3}	211,867	0.538	1.91
10^{-3}	29	9.74×10^{-4}	548,473	0.434	14.5	8.93×10^{-4}	291,463	0.438	1.88
$10^{-3.25}$	31.5	5.31×10^{-4}	554,183	0.433	17.5	4.61×10^{-4}	348,660	0.457	1.59
$10^{-3.5}$	37	2.94×10^{-4}	650,596	0.432	18.5	2.73×10^{-4}	377,874	0.453	1.72
$10^{-3.75}$	41.5	1.73×10^{-4}	738,357	0.432	19.5	1.52×10^{-4}	392,350	0.450	1.88
10^{-4}	46.5	9.77×10^{-5}	834,142	0.432	21.5	9.60×10^{-5}	433,217	0.451	1.93

D Time sweep

Table 5 lists the time sweep of Sec. 5.4 (Fig. 4).

Table 5: Time sweep at $(n_x, n_k) = (6, 8)$, target $\epsilon_{\mathrm{LCHS}} = 10^{-2.5}$ (paired rule at every t). Source: `runs/results_main.jsonl`, sweep `T`.

		ACL $\beta = 0.7$ (reference)				$\hat{f}_2$, QSVT (this work)				
t	$\|A\|t$	$k_{\max}$	error	STMC	succ.	$k_{\max}$	error	STMC	succ.	ratio
0.05	7.9	20.5	2.97×10^{-3}	215,773	0.496	5.5	2.70×10^{-3}	78,325	0.517	2.75
0.1	15.9	20.5	2.68×10^{-3}	384,688	0.438	7.5	2.62×10^{-3}	160,318	0.486	2.40
0.2	31.8	20.5	2.99×10^{-3}	730,155	0.364	8	2.98×10^{-3}	301,314	0.368	2.42
0.4	63.5	21	2.74×10^{-3}	1,425,405	0.286	6.5	1.94×10^{-3}	489,266	0.303	2.91
0.8	127.0	24	2.62×10^{-3}	3,216,781	0.215	6	2.24×10^{-3}	838,225	0.271	3.84

E Kernel-family characterisation

Tables 6 and 7 give the gate counts and success probabilities behind Fig. 5 and the three rankings of Sec. 5.5.

Table 6: STMC gates of the cheapest emulated circuit of each family that meets the target (standalone rule, $t = 0.1$, $n_x = 6$); "–": no $k_{\max} \leq 70$ reaches the target in the classical model (not emulated); $\times$: emulated but missed the target. Families: ACL $\beta = 0.7$ with brute-force weights and with QSVT–LCU weights, ACL $\beta = 0.9$ (QSVT–LCU), Cauchy (QSVT), $\hat{f}_2$ with $\alpha \leq 1.295$ (QSVT), $\hat{f}_2$ with $\alpha \leq 3$ (QSVT), $\hat{f}_{j,y}$ with LS's optimised parameters (QSVT–LCU). Last column: family with the fewest raw STMC gates. Source: `runs/results_main.jsonl`, sweep `C`.

n_k	ϵ_{LCHS}	$\mathrm{ACL}_{0.7}$	$\mathrm{ACL}^{\mathrm{Q}}_{0.7}$	$\mathrm{ACL}_{0.9}$	Cauchy	$\hat{f}_2$	$\hat{f}_2^{\alpha\leq 3}$	$\hat{f}_{j,y}$	fewest gates
5	$10^{-1.5}$	165,636	171,403	180,133	–	60,658	48,139	67,117	$\hat{f}_2^{\alpha\leq 3}$
5	10^{-2}	–	–	–	–	75,021	63,786	81,749	$\hat{f}_2^{\alpha\leq 3}$
5	$10^{-2.5}$	–	–	–	–	140,828	106,878	97,484	$\hat{f}_{j,y}$
5	10^{-3}	–	–	–	–	–	122,797	137,430	$\hat{f}_2^{\alpha\leq 3}$
5	$10^{-3.5}$	–	–	–	–	–	–	–	none
5	10^{-4}	–	–	–	–	–	–	–	none
6	$10^{-1.5}$	170,078	175,411	185,113	231,461	61,956	49,289	68,689	$\hat{f}_2^{\alpha\leq 3}$
6	10^{-2}	237,799	248,546	285,672	–	76,647	64,772	84,529	$\hat{f}_2^{\alpha\leq 3}$
6	$10^{-2.5}$	–	–	407,141	–	139,968	109,256	99,968	$\hat{f}_{j,y}$
6	10^{-3}	–	–	–	–	–	125,519	141,054	$\hat{f}_2^{\alpha\leq 3}$
6	$10^{-3.5}$	–	–	–	–	–	141,508	170,864	$\hat{f}_2^{\alpha\leq 3}$
6	10^{-4}	–	–	–	–	–	193,990	188,569	$\hat{f}_{j,y}$
7	$10^{-1.5}$	176,008	179,419	189,373	236,379	63,254	50,463	70,261	$\hat{f}_2^{\alpha\leq 3}$
7	10^{-2}	245,097	254,410	292,560	636,929	78,273	66,158	86,589	$\hat{f}_2^{\alpha\leq 3}$
7	$10^{-2.5}$	369,322	397,305	454,086	–	142,982	111,634	102,452	$\hat{f}_{j,y}$
7	10^{-3}	529,927	659,186	699,098	–	285,509	128,241	144,678	$\hat{f}_2^{\alpha\leq 3}$
7	$10^{-3.5}$	536,573	706,632	–	–	369,428	144,662	176,420	$\hat{f}_2^{\alpha\leq 3}$
7	10^{-4}	–	–	–	–	423,967	199,128	193,769	$\hat{f}_{j,y}$
8	$10^{-1.5}$	187,546	183,427	193,633	241,297	64,552	51,505	71,833	$\hat{f}_2^{\alpha\leq 3}$
8	10^{-2}	258,003	260,274	299,448	599,399	79,899	67,544	88,649	$\hat{f}_2^{\alpha\leq 3}$
8	$10^{-2.5}$	384,688	407,181	466,914	–	145,996	114,012	104,936	$\hat{f}_{j,y}$
8	10^{-3}	548,473	622,330	628,334	–	291,827	130,963	148,302	$\hat{f}_2^{\alpha\leq 3}$
8	$10^{-3.5}$	650,596	767,271	943,926	–	377,874	147,556	181,008	$\hat{f}_2^{\alpha\leq 3}$
8	10^{-4}	834,142	1,005,775	1,345,200	–	433,217	203,650	198,969	$\hat{f}_{j,y}$

Table 7: Success probabilities of the runs of Table 6. Last column: family with the smallest amplitude-amplified cost $N_{\text{gates}}/\sqrt{P_{\text{succ}}}$ (the gate count per success when the output is boosted by amplitude amplification).

n_k	ϵ_{LCHS}	$\text{ACL}_{0.7}$	$\text{ACL}_{0.7}^{\text{Q}}$	$\text{ACL}_{0.9}$	Cauchy	$\hat{f}_2$	$\hat{f}_2^{\alpha\leq 3}$	$\hat{f}_{j,y}$	cheapest $N/\sqrt{P}$
5	$10^{-1.5}$	0.47	0.50	0.22	–	0.45	0.23	0.33	$\hat{f}_2$
5	10^{-2}	–	–	–	–	0.45	0.27	0.27	$\hat{f}_2$
5	$10^{-2.5}$	–	–	–	–	0.50	0.08	0.23	$\hat{f}_2$
5	10^{-3}	–	–	–	–	–	0.08	0.20	$\hat{f}_{j,y}$
5	$10^{-3.5}$	–	–	–	–	–	–	–	none
5	10^{-4}	–	–	–	–	–	–	–	none
6	$10^{-1.5}$	0.50	0.50	0.22	0.80	0.45	0.23	0.33	$\hat{f}_2$
6	10^{-2}	0.46	0.46	0.20	–	0.45	0.27	0.27	$\hat{f}_2$
6	$10^{-2.5}$	–	–	0.19	–	0.47	0.08	0.23	$\hat{f}_2$
6	10^{-3}	–	–	–	–	–	0.08	0.20	$\hat{f}_{j,y}$
6	$10^{-3.5}$	–	–	–	–	–	0.09	0.19	$\hat{f}_{j,y}$
6	10^{-4}	–	–	–	–	–	0.08	0.17	$\hat{f}_{j,y}$
7	$10^{-1.5}$	0.49	0.50	0.22	0.80	0.45	0.23	0.33	$\hat{f}_2$
7	10^{-2}	0.45	0.46	0.20	0.76	0.45	0.27	0.27	$\hat{f}_2$
7	$10^{-2.5}$	0.44	0.44	0.19	–	0.47	0.08	0.23	$\hat{f}_2$
7	10^{-3}	0.44	0.44	0.19	–	0.44	0.08	0.20	$\hat{f}_{j,y}$
7	$10^{-3.5}$	0.43	0.44	–	–	0.45	0.09	0.19	$\hat{f}_{j,y}$
7	10^{-4}	–	–	–	–	0.45	0.08	0.17	$\hat{f}_{j,y}$
8	$10^{-1.5}$	0.50	0.50	0.22	0.80	0.45	0.23	0.33	$\hat{f}_2$
8	10^{-2}	0.45	0.46	0.20	0.77	0.45	0.27	0.27	$\hat{f}_2$
8	$10^{-2.5}$	0.44	0.44	0.19	–	0.47	0.08	0.23	$\hat{f}_2$
8	10^{-3}	0.43	0.44	0.19	–	0.44	0.08	0.20	$\hat{f}_{j,y}$
8	$10^{-3.5}$	0.43	0.44	0.18	–	0.45	0.09	0.19	$\hat{f}_{j,y}$
8	10^{-4}	0.43	0.44	0.18	–	0.45	0.08	0.17	$\hat{f}_{j,y}$

F Classical pre-processing cost

Table 8 summarises the classical cost discussed in Sec. 6.3.

Table 8: Classical cost per run (medians over all emulated runs of each weight construction; "/" separates median and maximum). Angle finding: QuCF's L-BFGS angle finder (single CPU core). Construction: QuCF circuit construction and I/O; emulation: QuEST state-vector evolution on one H200. Source: `logs/classical_cost.json`.

weight construction	runs	QSVT degree	angle finding [ms]	construction [s]	emulation [s]
brute-force weights (reference)	35	–	0.0 / 0.0	2.6	90.7 / 531.4
one even QSVT (real kernels)	68	24 / 96	1.2 / 18.6	2.3	22.8 / 305.6
two QSVTs + LCU (complex kernels)	71	32 / 474	2.3 / 869.3	2.5	73.3 / 2094.0